SITING EVIDENCE BUNDLE

Rural water point siting in Western Rural District, Sierra Leone

SUBJECT	Placement of 10 additional water points under a fixed budget
PREPARED BY	An autonomous analysis agent, from open data, without field verification
DECISIONS	All 7 decisions in this run were recorded by a person.
REVIEWED BY	Not reviewed. No planner overrides were applied to this run.
GENERATED	2026-08-27T02:55:17Z
RUN	`western rural-water-20260827T025305Z`
PROVENANCE	`03520a176445c3e9`

Statement

An estimated 1,520,809 people live in the area of interest. 160,241 of them, 10.5%, live within 1000 m straight-line walking distance of a water point recorded as serving. 1,360,568 people, 89.5% of the district, do not (Ex01, Ex02, Ex03).

Of the 4,382 water point records held for this district, 5 are recorded as serving and 4,377 are not (Ex01). The median record is 6.6 years old, so the figures above describe the surveyed state of the network rather than its state today (Ex10).

7,603 candidate locations were evaluated (Ex04). Selecting 10 of them by greedy coverage maximisation brings a further 576,307 people inside the service radius, raising coverage to 48.4% (Ex05, Ex06). Returns fall away quickly: the first site reaches 70,312 people and the last 44,478, so the number of sites to build is a decision that should be taken on the curve at Ex05 rather than on a round number.

No planner overrides were applied. The recommendation at Ex06 is the unmodified output of the selection method at Ex05.

2 of the 10 independent checks at Ex08 returned a flag rather than a pass: boundary effect, cartographic consistency. The flagged findings are reproduced in full at Ex08 and are not resolved in this bundle.

5 properties of the source registers would have changed a coverage figure had they been handled differently. Each is recorded at Ex10 with what was observed and what was done about it. None were corrected silently.

Recommendations in this bundle are advisory. Land availability, tenure, water table, community consent and construction feasibility are outside the data used here and inside the judgement of the officer reviewing it. No figure in this bundle is asserted without an exhibit number after it.

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Verification

Every retrieval in this bundle is recorded with its endpoint, the query as executed, the time of retrieval and the record counts before and after cleaning (Ex01, Ex02, Ex11). Cleaning rules are declared per source in the handbook files that accompany this bundle and are applied in order, each reporting its own drop count, so no record leaves the pipeline unaccounted for.

Coverage is recomputed independently of the solver by a separate process before this bundle is produced, and that process can prevent it being produced at all (Ex08). Coverage is always the union over serving locations; a per-location sum double counts every settlement reachable from two of them, and Ex08 reports what that sum would have claimed.

Properties of the source data that would change a figure if read differently are recorded at Ex10 rather than handled silently. Where the agent's own source handbook was found to be incomplete, that is recorded there too.

No figure in this bundle was estimated, interpolated or supplied from memory. A figure is either computed from a recorded retrieval or it is absent.

Ex00 – Decisions taken, and by whom

SUPPORTS That the judgements this assessment rests on are attributable
CAPTURED Recorded before the analysis ran
METHOD Each row is a judgement about values rather than a technical parameter: what counts as served, whose need the programme answers, whether a register of this age is fit to direct spending. The analysis refuses to run without them rather than assuming one, because assuming one would move the decision away from the officer accountable for it. Where a row is attributed to the agent, no district position was recorded and the agent decided in its absence.

All 7 decisions in this run were recorded by a person.

Decision	Value	Recorded by	Reason given
`service_radius_m`	1000 metres	transferability test	One kilometre, for comparability with the Uganda runs. Not a position held by any district in this country.
`coverage_basis`	straight_line	transferability test	Straight-line, to isolate whether retrieval and boundary matching transfer across countries without the friction surface as a second variable.
`objective`	max_coverage	transferability test	Maximum coverage, for comparability. Not a distributional position.
`budget`	10 facilities	transferability test	Ten, for comparability with the Uganda runs.
`data_currency_accepted`	test only, not fit for a real decision	transferability test	This run exists to check that the pipeline transfers, not to direct any spending. The register's age has not been assessed by anyone accountable.
`coverage_tolerance`	0.05 share of achievable coverage	transferability test	Five per cent, for comparability.
`equity_accepted`	unresolved	transferability test	No policy position is taken in a transferability test.

Ex01 – Rural water point siting register for Western Rural

SUPPORTS The installed stock and its recorded condition
CAPTURED https://data.waterpointdata.org/resource/eqje-vguj.json on 2026-08-27
METHOD Socrata query below, executed verbatim. Cleaning rules are declared in handbooks/wpdx.yaml and applied in order; each rule reports its own drop count.

```
clean_country_name = 'Sierra Leone' AND clean_adm2 = 'Western Rural'
```

Records returned	4,987
Records after cleaning	4,382
Discarded	605

RECORDS DISCARDED, BY RULE

flagged duplicate by WPdx	605

Recorded as serving	5
Recorded as not serving	4,377
Median record age	6.6 years

Ex02 – Population surface

SUPPORTS The denominator for every coverage figure in this bundle
CAPTURED https://hub.worldpop.org/rest/data/pop/wpgp on 2026-08-27
METHOD National 100 m grid, read over the area of interest and aggregated into blocks. The aggregation is coarser than the source and finer than the service radius, so it does not bias coverage.

```
SLE ppp 2020, bbox -13.2914 8.1440 -12.8749 8.5068 (WGS 84), aggregated 5x5
```

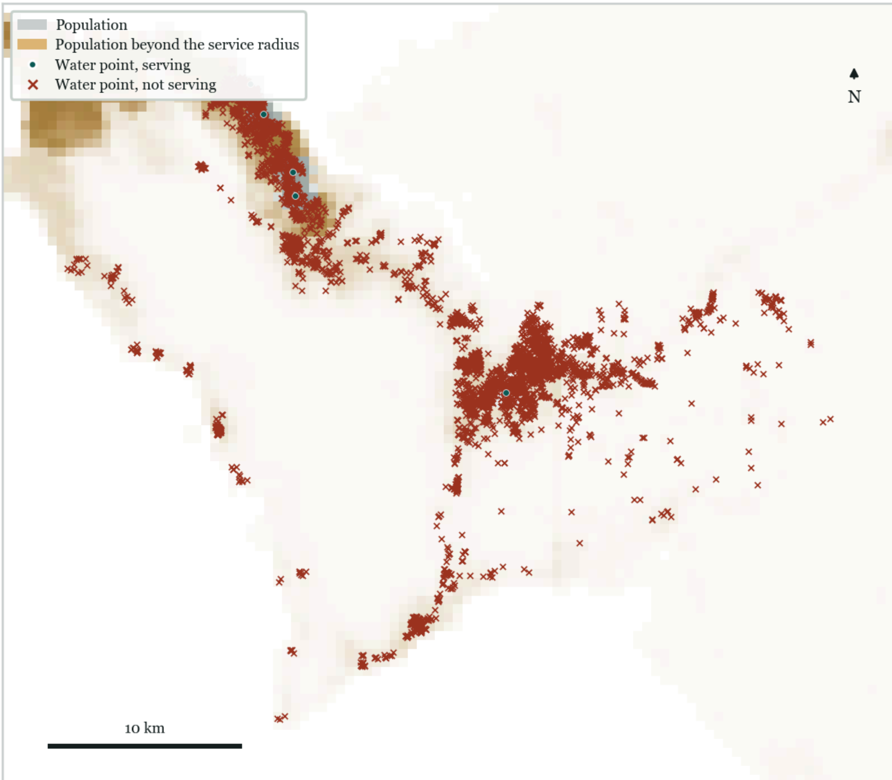
Raster cells in window	8,700
Aggregated demand cells	6,121
Population in the area of interest	1,520,809

population year 2020; 217,500 source cells at ~100 m read over the window, aggregated 5x5 into 8,700 cells of ~500 m

Ex03 – Coverage before any intervention

SUPPORTS The size of the gap the recommendation is trying to close
CAPTURED Computed from Ex01 and Ex02
METHOD A demand cell counts as covered when it lies within 1000 m straight-line walking distance of a point recorded as serving. Coverage is the union over serving points; a per-point sum would double count every settlement reachable from two of them.

Population	1,520,809
Covered today	160,241 (10.5%)
Not covered	1,360,568 (89.5%)



Serving and non-serving points over the population surface. Warm shading is population beyond the service radius.

Ex04 – Candidate sites considered

SUPPORTS That the recommendation was selected from a stated, reproducible set
CAPTURED Constructed from Ex01 and the area of interest
METHOD Two kinds of candidate: an existing point recorded as not serving, which could be rehabilitated, and a regular lattice of greenfield locations. Lattice points already inside an existing service area are removed before selection.

Candidates evaluated	7,603
Lattice spacing	750 m
Minimum spacing between recommendations	1000 m

Ex05 – Selection method and its guarantee

SUPPORTS That the ranking is reproducible and its optimality is bounded
CAPTURED Computed
METHOD Candidates are scored by the population they would newly bring inside the service radius and selected greedily. On a monotone submodular coverage function greedy selection is within a factor of 1 - 1/e, about 63 per cent, of the optimum. The exact integer programme is not used in the loop: it takes minutes at district scale where greedy takes seconds, and every planner override triggers a re-solve.

Sites selected	10
Population newly covered	576,307
Coverage after the programme	48.4%
Benchmark against an exact solver	not run (not run)

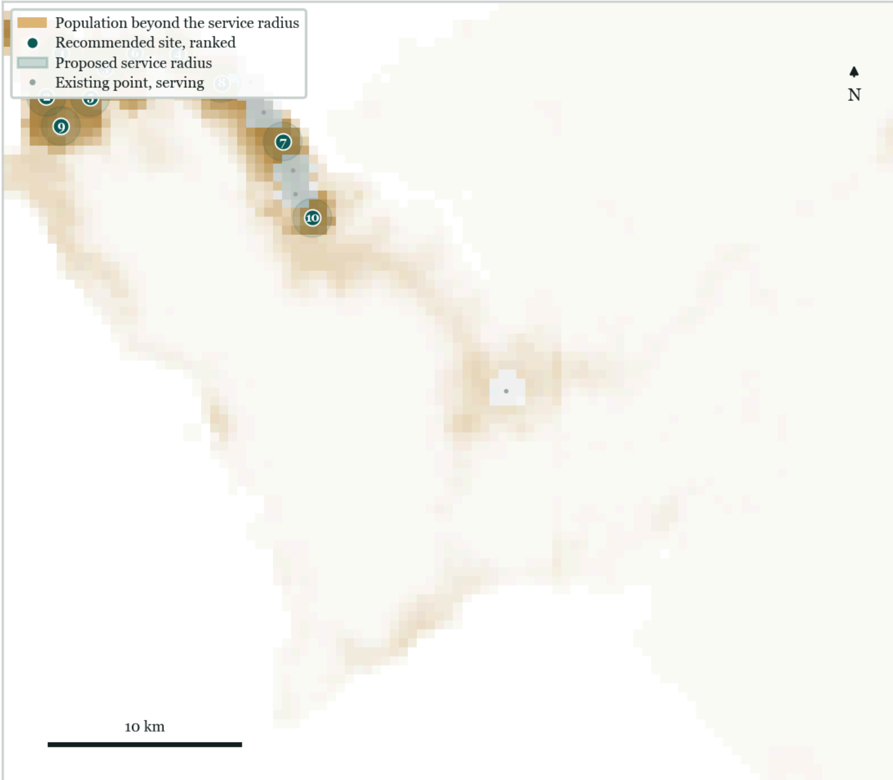
Sites built	Newly covered	Cumulative	Share of district
1	70,312	230,553	15.2%
2	69,414	299,967	19.7%
3	66,114	366,081	24.1%
4	61,985	428,066	28.1%
5	58,387	486,452	32.0%
6	55,023	541,476	35.6%
7	53,303	594,779	39.1%
8	52,772	647,551	42.6%
9	44,519	692,070	45.5%
10	44,478	736,548	48.4%

Marginal and cumulative population covered. The first site reaches 70,312 people, the last 44,478.

Ex06 – Recommended sites

SUPPORTS The recommendation itself, in coordinates
CAPTURED Computed
METHOD Sites in selection order. Each row's coverage is additional to every row above it, so the programme must be built from the top down. Coordinates are WGS 84.

No.	Site	Type	Location (WGS 84)	Newly covered	Nearest serving
1	S-001	greenfield	8.4829, -13.2643	70,312	9.8 km
2	S-002	greenfield	8.4626, -13.2712	69,414	10.5 km
3	S-003	greenfield	8.4761, -13.2439	66,114	7.5 km
4	S-004	greenfield	8.4827, -13.2098	61,985	4.0 km
5	S-005	greenfield	8.4625, -13.2507	58,387	8.3 km
6	S-006	greenfield	8.4828, -13.2302	55,023	6.2 km
7	S-007	rehabilitate	8.4419, -13.1607	53,303	1.6 km
8	S-008	greenfield	8.4690, -13.1894	52,772	1.5 km
9	S-009	greenfield	8.4490, -13.2644	44,519	10.0 km
10	S-010	rehabilitate	8.4063, -13.1469	44,478	1.5 km



Recommended sites and their service radii over the unserved population.

BASIS FOR EACH SITE

S-001 New site covering 70,312 people currently beyond the service radius; nearest working point 9.8 km away.

S-002 New site covering 69,414 people currently beyond the service radius; nearest working point 10.5 km away.

S-003 New site covering 66,114 people currently beyond the service radius; nearest working point 7.5 km away.

S-004 New site covering 61,985 people currently beyond the service radius; nearest working point 4.0 km away.

S-005 New site covering 58,387 people currently beyond the service radius; nearest working point 8.3 km away.

S-006 New site covering 55,023 people currently beyond the service radius; nearest working point 6.2 km away.

S-007 Rehabilitate the existing point covering 53,303 people currently beyond the service radius; recorded as functional, not in use; technology on record: Public Tapstand; nearest working point 1.6 km away.

S-008 New site covering 52,772 people currently beyond the service radius; nearest working point 1.5 km away.

S-009 New site covering 44,519 people currently beyond the service radius; nearest working point 10.0 km away.

S-010 Rehabilitate the existing point covering 44,478 people currently beyond the service radius; recorded as functional, not in use; technology on record: Hand Pump - Rope; nearest working point 1.5 km away.

Ex07 – Planner overrides and what they cost

SUPPORTS That the recommendation was reviewed by a person, and where it was overruled
CAPTURED overrides YAML, reproduced verbatim below
METHOD Each override was applied on its own and the plan re-solved, so every decision carries its own cost rather than one lumped figure. Overrides are never blocked. A veto removes an area, not a single lattice point, because a planner saying not there means a place.

No overrides were applied to this run. The recommendation in Ex06 is the unmodified output of Ex05.

Ex07b – Safety gate decisions

SUPPORTS That access, transfer and fallback decisions were taken deliberately
CAPTURED Recorded at the point of each decision

METHOD A gate never silently permits. Each row is a point at which the run either proceeded and recorded why that was safe, or refused. A refusal removes a source from the analysis and is stated here rather than left for the reader to infer from a missing figure.

Gate	Outcome	Decided by	Reason recorded
source authorisation	PASS	harness	Water Point Data Exchange Plus (WPdx+) is a public endpoint; no authorisation required
source authorisation	PASS	harness	WorldPop gridded population is a public endpoint; no authorisation required
guardrail breach	PASS	harness	Overrides forgo 0 people, 0.0% of the achievable coverage, within the 5% tolerance set for this run.

Ex08 – Independent checks

SUPPORTS That the plan was checked by a process that did not produce it
CAPTURED Computed
METHOD These checks run in a separate process from the one that produced the plan, can reject it, and do not read the override reasons in Ex07. They check the plan against the data, not against the account given of it. A single rejection stops the bundle being produced.

Check	Result	Finding
coverage arithmetic	PASS	union recount matches at 736,548; across the 10 new sites a per-facility sum would report 605,194 against a true union of 596,234, an overcount of 8,960
geometry	PASS	all 10 sites lie inside the area of interest
budget	PASS	10 sites within the budget of 10
coordinate reference system	PASS	EPSG:32628 (WGS 84 / UTM zone 28N); over 400 sampled pairs the projected distance departs from great-circle by 0.166% at the median and 0.532% at most, or about 5 m on a 1000 m service radius
data currency	PASS	median source record is 6.6 years old
aggregation sensitivity	PASS	reweighting the demand surface within cells moves the coverage claim from 48.4% to 48.5%, a shift of 0.10%
boundary effect	FLAG	the area of interest is an envelope around the retrieved records, not an administrative boundary; 0 of 10 sites lie within one service radius of its edge, nearest 2249 m. Demand and facilities beyond the envelope are unrepresented and the exposure is not well characterised
equity	PASS	100% of newly covered people live in the densest quartile of cells, against 95% district-wide
provenance	PASS	3 sources recorded, all figures traceable
cartographic consistency	FLAG	the rendered figures were not reviewed against the account; run the map-reviewer agent over figures.json and pass its verdict with --figure-review

Ex08b – Scoring review

SUPPORTS That the account was scored against a stated floor before issue
CAPTURED Computed by a reviewer that reads only this bundle's results file
METHOD The reviewer receives the results file and nothing else: not the solver, not the code path, not the override reasons. It cannot score the intention behind the plan, only the account given of it. Every dimension carries a floor and the weighted average carries a floor; the run is issued only when all are met, or when a person forces it and that is recorded.

Dimension	Score	Floor	Weight		Basis
data adequacy	6.5	5.0	0.25	met	median record 6.6 years old; 2 semantic anomalies in the register
method fitness	6.5	5.0	0.20	met	greedy maximum covering with a stated 1 - 1/e bound; no exact benchmark (not run); coverage is straight-line, not travel time
spatial rigour	8.0	6.0	0.25	met	6 of 6 required diagnostics present; distances in EPSG:32628 (WGS 84 / UTM zone 28N)
accountability	9.0	6.0	0.20	met	every figure resolves to a recorded retrieval; 5 source anomalies disclosed
actionability	9.5	4.0	0.10	met	10 sites with coordinates and a stated basis; marginal coverage curve supports a budget decision; 3 sensitivity scenarios

Weighted score	7.67
Floor to issue	6.50

Decision	issue
Reviewer	deterministic rules over results.json

Ex09 – Sensitivity of the recommendation

SUPPORTS How far the recommendation depends on assumptions the planner can change
CAPTURED Computed
METHOD Each scenario re-solves from the same data with one assumption changed. A recommendation that survives every row is one a planner can defend.

Scenario	Covered	Share	Against the base case
Rehabilitation of existing points only	422,497	27.8%	-314,051
Half the budget (5 sites)	486,452	32.0%	-250,096
Double the budget (20 sites)	1,008,757	66.3%	+272,209

Ex10 – Anomalies in the source data, and how they were handled

SUPPORTS That the source registers were read rather than assumed
CAPTURED Observed during retrieval and cleaning
METHOD Every item below is a property of the published data that would change a coverage figure if handled differently. Each is recorded with what was observed, what the agent did, and what a reader must keep in mind. None of these were corrected silently.

1. COVERAGE · GADM

Observed No administrative boundary could be matched for 'Western Rural' in SLE: "no unit named 'Western Rural' in SLE at levels (1, 2, 3). level 1: not found among 4 units; level 2: not found among 14 units; level 3: not found among 153 units. Closest by first
Handling The area of interest falls back to an envelope around the retrieved records, padded by a fixed margin.
Bearing Population from neighbouring districts may be counted in the denominator, and parts of the district with no recorded points may be excluded. Coverage shares are for the envelope, not the district.

2. SEMANTICS · WPdx+

Observed status_clean is not two-valued in this district. Values present: Non-Functional (3,234); Functional, not in use (1,137); Functional (4); Abandoned/Decommissioned (3); Non-Functional, dry season (3); Functional, needs repair (1).
Handling Serving status is read from the status_semantics block in handbooks/wpdx.yaml, which treats Functional and Functional, needs repair as serving and everything else as not serving.
Bearing An equality test against the string Functional would score every point in this district as not serving and inflate the gap.

3. SEMANTICS · WPdx+

Observed 3 records carry a status string not listed in the handbook.
Handling Treated as not serving, which is the conservative reading.
Bearing The gap may be overstated by up to these records.

4. CURRENCY · WPdx+

Observed The median record in this district is 6.6 years old (oldest 13.0 years, newest 0.6 years).
Handling Age is reported, not filtered. Uganda district holdings range from a mean of six to over sixteen years, so a staleness filter would delete whole districts rather than clean them.
Bearing Coverage figures describe the surveyed state of the network, not necessarily its state today.

5. DUPLICATION · WPdx+

Observed 605 of 4,987 returned records were removed: flagged duplicate by WPdx.
Handling Removed before any coverage computation, per the cleaning rules declared in the handbook.
Bearing Coverage computed on the raw response would count these records twice or count superseded observations.

Ex11 – Reproduction record

SUPPORTS That this bundle can be regenerated from the recorded queries
CAPTURED Computed
METHOD The bundle is a function of one results file and the manifest that produced it. Any change to a source query, a cleaning rule or a retrieval date changes the provenance hash.

Run	western rural-water-20260827T025305Z
Generated	2026-08-27T02:55:17Z
Provenance hash	03520a176445c3e9

```
python -m siting.cli --country Sierra Leone --adm2 Western Rural --iso3 SLE --decisions decisions/
generic.yaml --format bundle
```

Source	Licence	Accessed
GADM database of Global Administrative Areas, version 4.1, https://gadm.org	GADM licence, free for academic and non-commercial use	2026-08-27
WorldPop, School of Geography and Environmental Science, University of Southampton, https://www.worldpop.org	CC BY 4.0	2026-08-27
Water Point Data Exchange (WPdx+), global water point registry, https://www.waterpointdata.org/	CC BY 4.0	2026-08-27